

Square Integration

Connects **Square Appointments** with the Newo AI agent so callers and chat visitors can ask about services, book new appointments, and cancel existing ones — entirely by voice or chat, no receptionist required.

What the AI can do

- **Answer "do you offer X?" questions.** The agent searches the business's real Square catalog for any service the customer asks about, including variations, prices, and duration.
- **Check live availability.** Open time slots are pulled live from Square Bookings.
- **Book the appointment.** Once the customer picks a slot and gives their details, the agent creates the booking in Square — staff sees it immediately.
- **Cancel an appointment, even one made on a different day.** The agent recognises the returning customer by phone, fetches their upcoming bookings from Square, presents them, and cancels the one the customer points to.
- **Handle multiple upcoming bookings safely.** When the customer has more than one booking on file, the agent reads the list back and asks which one to cancel — never guesses.
- **Hide existing-client-only services from new callers.** Services flagged in Square as "for existing clients only" (e.g. pre-paid packages) are silently dropped from new callers' availability and consultation answers, with a polite explanation if asked by name and an offer to start with a non-gated alternative. Returning customers identified by phone see the full catalog. Works in chat too — the integration looks the caller up in Square mid-conversation as soon as a phone number is shared.
- **Route multi-service requests.** If a customer wants two or more services in one appointment, the agent politely hands off to a human manager.

Service catalog — how it syncs to the agent

Square is the single source of truth for services. The agent does **not** rely on the Products & Services list on the canvas — it queries Square directly.

When it happens	What the agent does
Every Publish All	<code>InitFlow</code> clears the last-sync timestamp and dispatches a fresh catalog fetch through <code>CatalogFlow</code> . The Square catalog (items with product type <i>Appointments Service</i>) is paginated for the selected location and the AKB is upserted.
First customer session after a deploy	<code>CatalogFlow/MaybeSyncCatalogOnSessionSkill</code> (subscribed to <code>session_started</code>) inspects the last-sync timestamp. If older than 2 hours (or empty), it kicks off a refresh — guaranteeing freshness even if no operator has hit Publish All.
New / renamed / re-priced service in Square	Hit Publish All in Newo for an immediate refresh, or wait for the 2h TTL to roll over on the next inbound chat / call.
Service deleted in Square	Same — next sync removes it from AKB.

Two coordinating attributes manage the cycle:

- **square_catalog_last_sync_at** — UTC timestamp of the most recent successful sync. Drives the 2h TTL gate on **session_started** . Cleared on every Publish All so the publish-time refresh always runs.
- **square_catalog_sync_in_progress** — UTC timestamp of an in-flight sync (used as a race lock). Cleared by the success / HTTP-error / platform- **runtime_error** paths. Stale locks older than 10 minutes are auto-taken-over so a crashed sync can never permanently block future syncs.

There is no operator-visible knob — the cycle is self-managing.

What the agent actually knows about each service

For every bookable variation the agent stores:

- **Service name** (parent item, e.g. *Eyebrow Threading*).
- **Variation name** (e.g. *Regular, Yearly Colour Refresh*).
- **Description** from the Square catalog (if any).
- **Price** — the fixed amount with its currency, or "variable" for variable-price services so the agent never quotes a wrong number.
- **Duration** in minutes.

Only bookable services reach the agent

Variations marked as "not available for booking" in Square, or restricted to a different location, are filtered out at sync time. The agent only ever offers and books what Square itself considers bookable right now.

Gating services to existing clients only

Some services should never be offered to new callers — typically pre-paid packages or course-of-treatments where the customer has already been seen and assessed. The integration supports a **marker phrase** in the Square service description that tags such services as existing-client-only.

How to mark a service. Open the service in Square Appointments and put the marker phrase anywhere in the description (case-insensitive). The default marker is **Only for existing client** :

Only for existing client
Pre-paid 5-session course of C02 Laser Resurfacing. Includes...

Operators can change the exact phrasing through the **Gated Services Marker** setting (**square_gated_services_marker**). Any phrase works; e.g. set it to **[Existing clients only]** and use that bracketed prefix in Square instead.

What the agent does. On every customer turn, the integration filters AKB matches whose **summary** **contains the marker** (case-insensitive substring match). The Square service description is embedded in the summary's **Description: ...** segment by the catalog sync, so the marker phrase ends up where the filter looks:

Caller status	What's filtered
Existing client (phone matched in Square)	Nothing — full catalog visible.
New caller (phone not in	Every gated service is dropped from availability/consultation results.

Square or never identified)	
Caller explicitly asked for a gated service	Agent gets a structured "not available for new clients" signal and replies politely: "That service is for existing clients only, but I can help with [non-gated alternative] instead — would you like that?" The block fires only when the caller's wording is specific enough to disambiguate the gated variation from non-gated ones — e.g. "Tattoo Removal Package". A generic parent term like "Tattoo Removal" silently falls through to LLM substitution against the non-gated variations instead of triggering the block.

Chat workaround. Voice channels auto-identify the caller by phone on conversation start. In chat there is no caller-id, so the integration runs the lookup mid-conversation: as soon as the customer shares a phone number, the agent quietly hits Square (`/v2/customers/search`), then re-triggers the availability check with a verified status. If the caller is in Square the gate lifts automatically; otherwise the gate stays in place.

Returning-customer recognition

When a phone number is detected on the conversation (caller-id on phone, or the customer mentions one in chat), the agent:

1. Searches Square customers by E.164 phone number.
2. Loads the customer's profile (name, email, address) into a private prompt section.
3. Pulls the customer's upcoming bookings (next 90 days) and stores them on the persona.

This means by the time the customer says *"please cancel my appointment"* the agent already has the booking in front of it — no searching, no awkward "what's your booking ID?" question. It also works **across sessions**: a customer who books today and calls tomorrow is recognised on the second call.

The behaviour is gated by **Existing Client Lookup** ; turn it off if you don't want a Square API call per session.

Features at a glance

Feature	Included
Answer questions about any service from the Square catalog	✓
Quote prices and durations from the live catalog	✓
Check live availability	✓
Book appointment	✓
Cancel appointment (current session)	✓
Cancel appointment created on a different day	✓
Auto-recognise returning customer by phone, pre-load their bookings	✓

Hide existing-client-only services from new callers (configurable marker)	✓
Re-verify caller in chat when phone shared mid-conversation	✓
Disambiguate multi-booking cancel — agent asks which one	✓
Auto-create customer profile in Square during booking	✓
Reuse existing Square customer when phone matches	✓
Assign the right team member to each slot	✓
Multi-service request — transfer to a human	✓
Two auth methods: Personal Access Token (simple) or OAuth 2.0	✓
Reschedule in one step	✗ (cancel + rebook)
Take payments in Square	✗ (Square handles payment at the appointment as normal)

Scenarios

The integration ships with three editable scenarios on the canvas. Operators can tweak them in Newo; the agent picks up the changes on the next publish.

Scenario 9 — Service Consultation in Square

Runs when the customer is asking about a service without yet asking to book.

- Identify what the customer is asking about. The scenario branches in four ways:
 - **Specific service** ("eyebrow threading", "Hydra Facial") → straight to catalog lookup.
 - **Body area or category** ("what for my legs?", "popular brow options", "something for hair removal") → catalog lookup using the area as the search hint. The agent never gets stuck on "tell me a specific name".
 - **Comparative follow-up** ("which is cheapest?", "what's the longest one?", "what's the difference between 1 session and 3?") about variations the agent **just** listed → answered directly from the data already in the previous turn. **No second catalog call.**
 - **Fully generic** ("what services do you offer?") → polite ask for an area first.
- Look up the catalog** — agent says *"Let me check our services for you, I will get back to you shortly."*
- Present matches with **price and duration** so the customer can compare on the spot:
 - Found → confirm the service is offered, list up to N variations (configurable, default 3), quote price and duration each, invite the customer to book.
 - Not found → politely decline and invite the customer to describe what they need so the agent can suggest a related service.
- If the customer wants to book, hand off to **Scheduling Appointment in Square via Agent**; otherwise wrap up.

Anti-hallucination guard. The agent must never voice a catalog service name that did not come from a recent consultation tool result. If the customer asks "what's popular?" before any catalog lookup has run, the agent first asks which area, then runs the lookup, then answers — it never invents service names from generic spa knowledge.

Triggered by intent "[T] Service Consultation in Square".

Scenario 7 — Scheduling Appointment in Square via Agent

Runs when the customer wants a new appointment. Each data-gathering step delegates to a shared NAF procedure (Gathering Preferred Date and Time , Reconfirming or Gathering Phone Number , Reconfirming or Gathering First and Last Name) so the agent reuses the same channel-aware logic as the rest of the system.

1. **Service Identification.** Ask what service the customer wants. **As soon as a single service is named the agent moves on** — availability is the source of truth.
2. **Capture preferred date AND time** via the **Gathering Preferred Date and Time** procedure. The procedure reconfirms vague phrases ("any day", "tomorrow afternoon") explicitly before the agent commits to a value. The availability code-phrase is forbidden until both date AND time are captured.
3. **Capture phone number** via the **Reconfirming or Gathering Phone Number** procedure. **Collecting phone BEFORE availability is intentional:** it lets the integration do an existing-client lookup against Square (/v2/customers/search) and lift the existing-client gate (see §Gating below) for callers who turn out to be in Square. Channel-aware:
 - **Phone channel** — the procedure sees the caller-id from <UserInformation> and just asks permission to use the existing number.
 - **Chat channel** — the procedure asks the User to share a phone number when none is on file.
4. **Check live slots** — agent says *"Let me check available time slots for you, I will get back to you shortly."*
5. Read the <AvailableSlots> block:
 - **Slot matches the captured preference** → confirm in one short line: *"Yes, [date] at [time] is available — would you like me to book it?"*.
 - **Slot doesn't match** → offer 2–3 nearest alternatives from the same day (or week if the day is empty) verbatim from <AvailableSlots> ; ask the customer to pick.
 - **Loading** → wait, give a brief answer to anything the customer asks meanwhile.
 - **Error** → during working hours, transfer to a human; outside hours, relay a message to the manager.
 - **Multiple-match disambiguation** (status="needs_clarification") → list the candidates the agent received, ask the customer to pick. After they pick, repeat step 4 (re-say the code-phrase) so the tool fires again with the clarified service.
 - **Existing-client-only block** (status="not_available_for_new_clients") → the requested service is reserved for existing clients **and** the phone we have on file is not in Square. Politely tell the caller, offer a non-gated alternative we DO offer to new callers, and do NOT proceed to confirmation. If the caller claims existing-client status under a different phone number, return to step 3 to recapture phone, then re-fire availability via step 4.
 - **No catalog match** (status="no_match") → ask the caller to specify the service more precisely and return to step 1.
6. **Customer Name** via the **Reconfirming or Gathering First and Last Name** procedure. Skipped automatically if first and last name are already on file from earlier in the conversation.
7. **Read back the booking** — service, day, time — and ask *"Should I go ahead and book [service] on [date] at [time]?"*. On confirm, proceed to step 8.

8. **Submit the booking** — agent says "Give me a moment, I will book your appointment right now." → booking is created in Square.
9. If the customer asks for **two or more services** at any point, the agent transfers to a human manager.

Phone is taken **exactly as captured** in E.164 format (e.g. +447426906344); domestic formats like 07426906344 are not accepted by Square.

Triggered by intent "[L] Appointment in Square via Agent".

Scenario 8 — Canceling Appointment in Square via Agent

Runs when the customer wants to cancel.

1. Agent says "Give me a moment to double-check if I have information about your bookings, I will get back to you shortly." The integration has already populated the <ExistingBookings> prompt section either from the current-session booking, the persona cache, or a fresh phone lookup against Square.
2. **Read <ExistingBookings>** :
 - **Empty** → during working hours, transfer to a human; outside hours, relay a message to the manager.
 - **One booking** → present its date/time, ask "are you sure?".
 - **Multiple bookings** → list them and ask which one to cancel. Once the customer points at one, the integration uses an LLM extraction validated against the actual booking_id list to pick the right one.
3. Customer confirms → agent says "Give me a moment, I will cancel your booking right now." → cancellation confirmed in Square.

Triggered by intent "[T] Cancellation in Square via Agent".

Each scenario's exact code-phrase is required — it is the trigger that signals the agent to actually call Square. If you rewrite a scenario, keep those phrases intact.

Example dialogs

Example 1 — Consultation

Customer: Tell me about microblading, what are my options? **Agent:** Let me check our services for you, I will get back to you shortly. **Agent:** Yes, we offer microblading. We have Microblading for £300, 90 minutes, and Microblading Yearly Colour Refresh for £180, also 90 minutes. Which one would you like?

Example 2 — Direct booking (no consultation step)

Customer: Hi, I'd like to book Microblading Yearly Colour Refresh. **Agent:** Let me check available time slots for you, I will get back to you shortly. **Agent:** I have Friday May 1 at 12:45 PM, 2:15 PM, and 3:45 PM. Which works? **Customer:** 2:15 PM. **Agent:** May I have your name, phone, and email? **Customer:** Sophia Reyes, +447426906344, sophia.reyes@example.com. **Agent:** Give me a moment, I will book your appointment right now. **Agent:** You're all set, Sophia — your appointment is confirmed for Friday May 1 at 2:15 PM.

Example 3 — Cross-session cancel

Customer calls back the next day to cancel the booking from Example 2.

Customer: Hi, I need to cancel my appointment. My phone is +447426906344. **Agent:** Give me a moment to double-check if I have information about your bookings, I will get back to you shortly. (Agent looks up the customer in Square, finds their upcoming booking.) **Agent:** I found one upcoming booking for you on Friday May 1 at 2:15 PM. Are you sure you want to cancel? **Customer:** Yes, cancel it please. **Agent:** Give me a moment, I will cancel your booking right now. **Agent:** Your appointment for Friday May 1 at 2:15 PM has been successfully cancelled.

Example 4 — Multi-booking cancel

Customer: Hi, please cancel my appointment. My phone is +447426906344. **Agent:** Give me a moment to double-check if I have information about your bookings, I will get back to you shortly. **Agent:** I found two upcoming bookings: April 28 at 2:15 PM and April 30 at 5 PM. Which one would you like to cancel? **Customer:** April 30 at 5 PM please. **Agent:** Are you sure you want to cancel your booking for April 30 at 5 PM? **Customer:** Yes. **Agent:** Give me a moment, I will cancel your booking right now. **Agent:** Your booking for April 30 at 5 PM has been successfully cancelled.

Example 5 — Multi-service → manager

Customer: I'd like to book microblading and lip filler in the same visit. **Agent:** For appointments with multiple services, I'll need to connect you with our manager so they can arrange everything together. One moment...

Example 6 — Policy question (not a service lookup)

Customer: Can I come drunk to my appointment? **Agent:** We recommend avoiding alcohol for 24 hours before and after your treatment.

Policy questions are answered from the Business Context on the canvas, not from the catalog.

Setup

Create the Newo project

If this project already exists in **Newo Builder**, skip this subsection and continue with the integration-specific settings below.

1. In **Newo Builder**, open the projects list and click **Create Project** (or **Create New Project** from the top-right menu).



Create New Project menu in Newo Builder

2. Fill **IDN** and **Title**. The exact names do not matter; use any clear names your team will recognize.
3. In **Registry**, choose the release channel:
 - **staging** — the newest module fixes appear here first. Use it when you need the latest fix, but expect possible unfinished changes.
 - **production** — the final stable version for live projects.
4. In **Module**, select the module for this integration.



Create Project form showing IDN, Title, Registry, and Module fields 5. Leave **Module version** on **Latest**

version unless support tells you to pin a specific version, then click **Create**.

A step-by-step walkthrough with screenshots is in [Square Onboarding Guide \(PDF\)](#). The section below is the short version.

1. Prepare your Square account

1a. Add a location

Open **Square Dashboard** → **Locations** and make sure at least one location exists. Set its address, timezone, and business hours.

My Business
Square Free [See Plans](#)

Search

Settings

Account & Settings

Personal information

My business

About

Security

Locations

Pricing & subscriptions

Payments

Banking

Notifications

Tax forms

CCPA

Fulfillment methods

Order tickets

Take payment

Locations

Location filter

Create Location

1 location is active for payments but does not have an address. Please update its address.

Nickname	Location Type	Address	City, State	Phone	Email
My Business	Physical	Update Address			morenko83+test@gmail.com

![[Location details]](screenshots/location details.png)

1b. Turn on Square Appointments for the location

Open **Square Appointments** → **Settings** and enable Appointments for the location.

My Business
Square Free [See Plans](#)

Search

Home

Items & services

Payments & invoices

Online

Customers

Reports

Staff

Banking

Settings

Add more

Take payment

Add more to Square

Customize Square for your unique business needs, with enhanced features for selling, managing your business, and reporting.

Inventory **Bookings** Restaurants

Streamline appointment bookings and protect against no-shows. [Add](#)



Popular features

Appointment scheduling

No-show protection

Combined services and item checkout

Online booking calendar

Automated client reminders

Daily appointment limits and waitlists

Learn more

Explore all features >

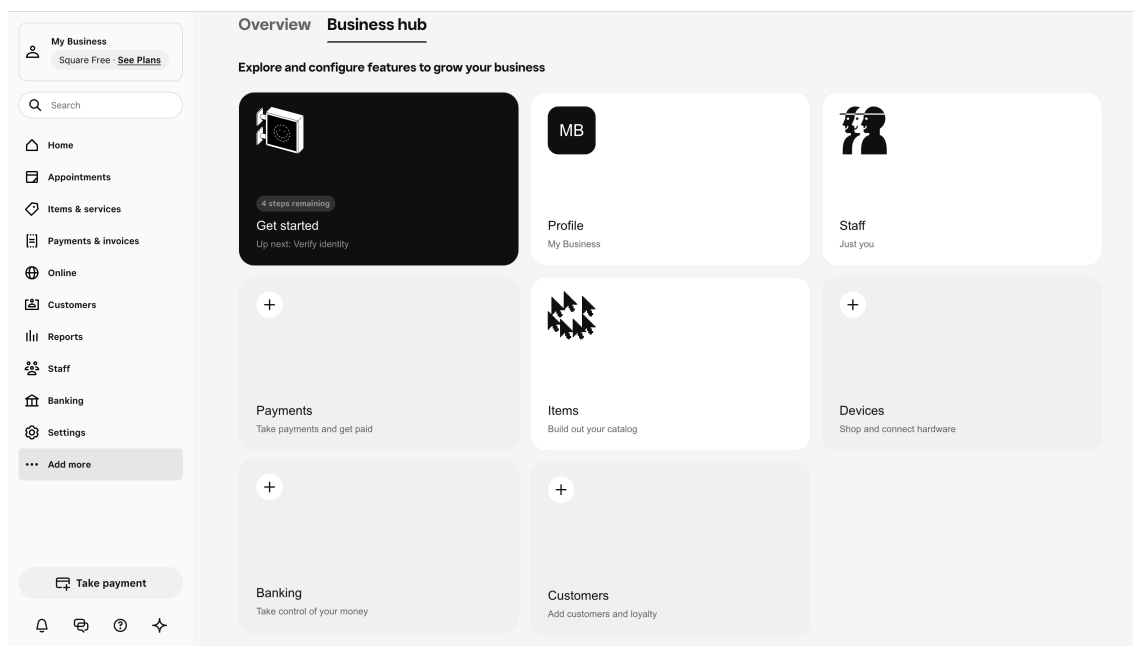
×

Start setup

Scheduling features were successfully added

Use the setup guide to start using these features.





1c. Add at least one team member who can take bookings

My Business

Square Free · [See Plans](#)

Q

Search

←

Staff

Team

Team members

Permissions

Onboarding

Scheduling

Time tracking

Payroll

Announcements

Settings

Take payment

🔔

🗨

🔗

🌟

Team members

Q

Search

Job

Location

Export

Add team member

Name ↑	Job	Location	Appointments	Status
👤 Owner	Owner	All current and future locations	-	Incomplete

1 team member

×

Next

1d. Define services

Open **Square Appointments → Services** and add every service the agent should offer. Each service must have a price and duration; tick **Available for booking** on every variation the agent should offer; use **VARIABLE_PRICING** only when the price genuinely is variable.



Create service

Save

Name

Lip Filler

Li

Edit

Description

Cosmetic injections that add volume, shape, and definition to the lips

Generate

Drag and drop images here, [upload](#), or [browse image library](#).

Take your service photos to the next level.

Scan to download the Photo Studio app for free. [Learn more](#)

App Store

Locations

twinmind

Categorization

Categories

Select



Edit service

Delete

Save

Price

Add variation images

Price type

Fixed

Price

\$100.00

Cancellation fee has moved to Payments & Cancellations settings. You can change it by going to Appointments > Settings > Payments & cancellations.

[Go to settings](#)

Cost

Track service cost

Track the cost of providing this service (e.g. back bar cost) and allow service costs to be deducted in **Commissions**.

Duration

Hours

0 hrs

Minutes

30 mins

Add processing time

Optimize your schedule and allow customers to book during processing time.

Block extra time after appointment

Avoid rushing between appointments by blocking off time for cleanup or travel.

X

Edit service

DeleteSave

Add processing time★

Optimize your schedule and allow customers to book during processing time.

Block extra time after appointment

Avoid rushing between appointments by blocking off time for cleanup or travel.

Online booking

Bookable by customers online

Allow customers to book this service on your booking website.

ⓘ

Assigned team members

All team members

Select

Replace price with description

Request that clients inquire about pricing for appointments where rates flex or differ.

Variations

Create multiple variations of a single service. Set different pricing options, durations, and more.

Add

Advanced settings

Set-up modifiers and custom attributes

▼

My Business

Square Plus 30 days left

Search

Items & services

Items

Item library

Channel listings

Service library

Packages

Image library

Resources

Modifiers

Categories

Discounts

Options

Units

Custom attributes

Settings

Take payment

Search

All locations

Actions

Create service

Name	Reporting category	Duration	Price
Consultation (example service)		1 hr	Variable
Lip Filler		30 mins	\$100.00

1e. Confirm availability

Open **Square Appointments → Calendar / Availability** and make sure each team member has working hours.

My Business

Square Plus

30 days left

Q

Search

←

Staff

Team

▼

Scheduling

▲

Schedule

Availability

Time off

Time tracking

▼

Payroll

Announcements

Settings

▼

Availability

★

twinmind

▼

Team member	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Owner							
Test Test							



Monday



Available all day

Start Time

9:00 AM ▼

End Time

5:00 PM ▼

Add availability.

Repeat on



Sunday



Monday



Tuesday



Wednesday



Thursday



Friday



Saturday

Save

Availability ★

twinmind ▼

Team member	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Owner							
Test Test ▼		Available 9 am – 5 pm	Available 9 am – 5 pm	Available 9 am – 5 pm	Available 9 am – 5 pm	Available 9 am – 5 pm	



Next

Jobs

Manage Test's role and compensation.

Primary job	Cashier, Barista, Bartender, etc.
Pay type	Hourly ▼
Hourly rate	\$0.00

Add another job

☐

Overtime Exempt

This team member is not exempt from overtime FLSA rules. [Learn more](#).

2. Get your Square credentials

Option A — Personal Access Token (recommended)

- 1. Go to <https://developer.squareup.com/apps>.
- 2. Create a new application, or use an existing one.
- 3. Open the app → **Credentials** → **Production** tab.
- 4. Copy the **Personal Access Token**.

[Next](#)

Access

Manage what Test can see and do across Square.

Assign permissions

Permission set
Team Permissions

[Edit Team Permissions](#)

Locations

2 Locations

Authorized representative

Allow Test to contact Square Support on behalf of your business. [Learn more](#)



Create a passcode for Test



Using this passcode or a [team member badge](#), Test can log in to the shared POS with the permissions you have assigned.

4 7 3 4

Option B — OAuth 2.0

1. In the same Developer Dashboard app, copy the **Application ID** and **Application Secret**.
2. During setup you'll be redirected to Square to authorize the app and paste the resulting code back into Newo.

3. Configure in Newo

1. Open the project and find the **Module - Square Integration** attribute group.
2. Choose auth method in **square_auth_type** (**Token** or **OAuth**).
3. For **Token**: paste the token into **square_personal_token** . Hit **Publish All**.
4. For **OAuth**: fill in **square_client_id** and **square_secret_id** , follow the Square authorize link, paste the returned code into **square_oauth_code** , then **Publish All**.
5. After the first successful auth, **select the location** in the **square_locations** dropdown (auto-populated) and hit **Publish All** again.
6. (Optional) review the feature toggles:

Setting	Purpose	Default
Availability Feature	Live slot lookup on/off	on
Booking Feature	Booking creation on/off	on
Cancellation Feature	Cancellation on/off	on
Service Consultation Feature	"Do you offer X?" / "How much is Y?" lookup	on
Existing Client Lookup	Auto-recognise returning customer by phone, pre-load their Square bookings on every conversation start	on
Auto-Setup Canvas Scenarios	Publishes the consultation/booking/cancel scenarios on first install	on

4. Hit Publish All

On publish, the agent will:

- Create the Square connector.
- Register the custom tools (consultation, availability, booking, cancellation, search bookings).
- Pull the Square catalog for the selected location and load it into the agent's knowledge base.
- Add the consultation / booking / cancellation scenarios and intents to the canvas (existing customizations are preserved).

Place a test chat message or call to confirm.

AKB ↔ Square Catalog Sync

The agent's *Service Consultation* and *Availability* tools never call Square mid-conversation to "find" a service — that would be too slow on a live call. Instead, the integration mirrors the Square catalog into the agent's **AKB (Active Knowledge Base)** as a set of topics that the agent can fuzzy-match against in milliseconds.

What ends up in AKB

For every bookable variation in Square, the integration writes one AKB topic on the **SquareupAgent persona** with:

AKB field	Value
Name	<service_name> – <variation_name> (e.g. Brow Lift – Regular)
Summary	Human-readable blob: Service: <name> – Variation: <name> (ID: <variation_id>) Price: 120.00 GBP Duration: 15 min Description: ...
Facts	Structured key/value list: service_name, variation_name, variation_id, price, duration. Service description lives only in Summary — gating filters substring-match against it directly.
Source	square_catalog
Labels	square_catalog
Confidence	90

The label is what scopes every read — both the consultation tool and the availability tool query AKB with `labels=["square_catalog"]` , so other AKB topics on the persona are never confused with Square data.

When the sync runs

Trigger	What happens
Every Publish All in Newo	InitFlow → SetupSkill emits square_update_catalog after the integration is initialised. CatalogFlow paginates Square Catalog API → /v2/catalog/list?types=ITEM for the selected location, filters by product_type=APPOINTMENTS_SERVICE and available_for_booking=true, then rewrites the AKB.

Service added / renamed / re-priced in Square	Hit Publish All in Newo. The new state is live on the next customer turn.
Service marked unavailable / deleted in Square	Same — publish and the topic is removed from AKB.
Square webhook (catalog.version.updated)	Not wired in this version. Operators must publish to pick up changes.
Scheduled re-sync	Not implemented — the publish-on-demand model is intentional so the operator controls when the catalog goes live.

How duplicates are handled

The sync is an **upsert** — never a wipe-and-rebuild. AKB stays consistent during the refresh, so a customer asking about availability while the catalog is being re-synced still gets a real answer instead of "we don't offer that".

`_saveCatalogToAkbSkill` does the following on every publish:

1. Loads existing `label=square_catalog` topics and builds a `variation_id → topicId` map from their Facts.
2. For each variation in the fresh Square pull:
 - **variation_id already in AKB** → `UpdateAkb` rewrites the topic in place (new price, duration, description, name).
 - **New variation_id** → `SetManualAkb` creates a fresh topic.
3. After every variation has been upserted, **only then** does it `DeleteAkb` the orphans — topics whose `variation_id` is no longer in Square — plus any legacy rows whose Facts don't carry a parseable `variation_id`.

There is no moment when AKB is empty. The consultation and availability tools can fire safely throughout the refresh.

Duplicate variation names across different services are still kept separate — `Styling - Fringe Cut` and `Styling - Ladies Wash, Cut and Finish` are distinct topics because their AKB **Name** is `<service> - <variation>` and the **Facts** carry the unique `variation_id`. The agent disambiguates via `variation_id` whenever it books or quotes a price, so two topics that share a `variation_name` (e.g. two services that both have a *Regular* variation) never collide.

How stale records are removed

There is no soft-deprecate / archive flag. Each publish reconciles the AKB against the fresh Square pull and deletes any topic whose `variation_id` is no longer in Square (or whose Facts are corrupt / legacy). **Any service no longer present (or no longer bookable) in Square is gone from AKB after the next publish.** The consultation tool will then answer "we do not offer that" and offer to suggest a related service, exactly as if the service had never existed.

How the agent uses these records

Tool	How it queries AKB
------	--------------------

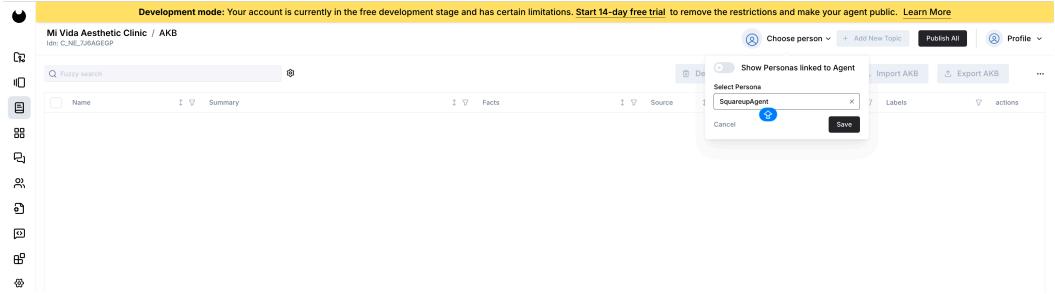
Service Consultation (<code>_serviceConsultationSkill</code>)	<code>SearchFuzzyAkb(query=<customer's words>, searchFields=["name", "summary"], labels=["square_catalog"])</code> — match width and strictness come from Catalog max topics (default 10) and Catalog score threshold (default 0.1). The reply to the customer is then capped to Consultation max services (default 3) so voice channels don't enumerate ten options.
Availability (<code>_checkAvailabilitySkill</code>)	<code>SearchFuzzyAkb(...)</code> with the same two attributes. The agent then either picks one variation (if a single match is unambiguous) or asks the customer to disambiguate before checking slots — see the disambiguation case in Scenario 7.
Booking	Uses the <code>variation_id</code> already resolved by the availability step — no fresh AKB lookup.

The fuzzy match is forgiving: typos, partial names, and reordered words still resolve to the right topic as long as the score clears the configured threshold. Operators can tighten the threshold (e.g. `0.3`) or widen the candidate cap (e.g. `20`) without a redeploy by editing the `square_akb_*` attributes and republishing.

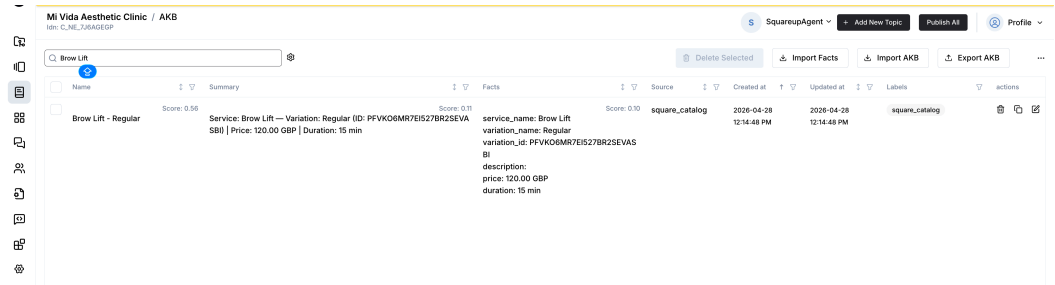
Verifying the catalog in Newo (AKB search via UI)

Operators can inspect what the agent actually sees at any moment:

1. Open your project in Newo and click the **AKB** icon in the left navigation.
2. In the top-right corner, click **Choose person**, toggle on **Show Personas linked to Agent**, and select **SquareupAgent** in the *Select Persona* dropdown. Click **Save**.



3. The grid is now scoped to the SquareupAgent's knowledge base. Type a service name into the **Fuzzy search** box (e.g. `Brow Lift`) and press Enter. Matching topics appear with their **Summary**, **Facts**, **Source** (`square_catalog`), and **Labels** (`square_catalog`).



The **Score** next to Name / Summary / Facts is the fuzzy-match score for the query — the agent uses the same scoring on every customer turn.

This view is the source of truth when troubleshooting: if a service is missing here, it is missing for the agent.

Troubleshooting

Symptom	Likely cause	Fix
AKB is empty after Publish All.	Auth not yet configured (token / OAuth code missing or invalid) — the catalog skill bails out early and logs [CATALOG] Skipping catalog sync — no auth configured yet.	Finish the auth step in Setup, then Publish All again.
AKB is empty but auth works.	Square location has zero items with <code>product_type=APPOINTMENTS_SERVICE</code> , or every variation is marked <i>not available for booking</i> .	In Square: confirm the items are <i>Appointments services</i> (not <i>Retail</i>) and tick Available for booking on at least one variation. Republish.
Some services missing after Publish All.	The variation isn't bookable for the selected location (<code>present_at_location_ids</code> filter), or <code>available_for_booking=false</code> .	Adjust the variation in Square (Locations + Available for booking), republish.
Price or duration look wrong in AKB.	Catalog in Square was edited but operator did not republish.	Publish All in Newo. AKB is a snapshot of Square at publish time.
Price shows variable.	Variation uses <code>pricing_type=VARIABLE_PRICING</code> in Square.	Intentional — the agent will not quote a fake price. Set a fixed price in Square if you want a number quoted.
Two near-identical AKB rows for the same service.	Two different bookable variations with the same <code>variation_name</code> exist in Square.	Expected — the rows differ by <code>variation_id</code> in Facts. The agent picks the right one via <code>variation_id</code> at booking time, not by name.
Old (deprecated) service still appears in AKB.	Operator removed it from Square but did not republish.	Publish All . The upsert step removes any topic whose <code>variation_id</code> is no longer in the fresh Square pull (and any legacy row without a parseable <code>variation_id</code>).
Fuzzy search in the UI returns 0 hits but the topic exists.	The persona filter in the top-right is wrong (e.g. left on a previous agent).	Re-open Choose person , pick SquareupAgent , Save , retry the search.

Customizing the agent

The scenario bodies on canvas are fully editable. Common tweaks:

- **Ask for more info during booking.** Add steps like "ask for the allergy history" or "ask for preferred staff".
- **Change the consultation tone.** Scenario 9 can be rewritten — just keep the code-phrase *"Let me check our services for you, I will get back to you shortly."* intact so the lookup still fires.
- **Restore defaults.** If you break a scenario, delete it from the canvas and republish — the original is restored from the integration's library.

Frequently asked questions

Q. Does the agent ever charge the customer? No. It only books and cancels. Payment is handled inside Square by staff as normal.

Q. What happens if two customers try to book the same slot? Square serializes the bookings. The second request comes back with a "slot no longer available" error; the agent apologizes and offers the remaining slots.

Q. Can the caller reschedule in one step? Not directly — the agent cancels the old booking and starts a new booking flow.

Q. The customer asked about a service we stopped offering last week. After the next **Publish All**, the old service is gone from the knowledge base — the consultation scenario will tell the customer it's not offered and invite them to suggest a related service.

Q. We just added a new service in Square. When does the agent see it? Hit **Publish All** in Newo. The agent already knows about the new service on the next customer turn.

Q. Does the agent answer policy questions? Yes — those come from the **Business Context** on the canvas, not from the Square catalog.

Q. Can I connect multiple Square locations at once? One location per integration instance.

Q. What happens if the token expires or is revoked? *Token mode:* tokens don't expire but can be revoked. Generate a new one and update `square_personal_token`. *OAuth mode:* the integration auto-refreshes on 401 using the refresh token — no manual action needed.

Q. Can I turn off the consultation tool? Yes — set **Service Consultation Feature** to off and republish.

Q. Can I turn off the existing-client lookup at conversation start? Yes — set **Existing Client Lookup** to off. The agent will still be able to find bookings on demand via the search-bookings tool when the customer asks to cancel.

Q. Why E.164 phone numbers only? Square's customer search and booking endpoints reject domestic formats (e.g. UK `07426906344`). The integration insists on the international format (`+447426906344`) so the API accepts the request.

Limitations

- **Reschedule** endpoint is not wired (Square API shape differs per region). Workaround: cancel + rebook.
- **Multi-service bookings** are intentionally routed to a human manager.
- **Historical data** — the agent only sees services and bookings that exist in Square at or after the first publish.
- **One location per instance** — to serve two locations, deploy two Newo projects.

All settings reference

Every attribute the integration adds to the **Module - Square Integration** group in Newo. Operators typically only touch the "Primary" rows; the rest are good out of the box.

Primary (set these during setup)

Attribute	Title in UI	Purpose	Default
square_auth_type	Authentication Method	Token (paste Personal Access Token) or OAuth (go through Square authorize flow).	Token
square_personal_token	04. Personal Access Token	The token from Square Developer Dashboard (Token mode).	(empty)
square_client_id	02. Application ID	OAuth application ID (OAuth mode).	(empty)
square_secret_id	03. Application Secret	OAuth application secret (OAuth mode).	(empty)
square_oauth_code	01. OAuth Code	One-time authorization code returned by Square (OAuth mode).	(empty)
square_locations	Square Location	Dropdown of locations. Pick the one this integration instance serves.	(empty)
square_feature_availability_enabled	Availability Feature	Master on/off for live slot lookup.	on
square_feature_booking_enabled	Booking Feature	Master on/off for creating bookings.	on
square_feature_cancellation_enabled	Cancellation Feature	Master on/off for cancelling bookings.	on
square_feature_consultation_enabled	Service Consultation Feature	Master on/off for the catalog lookup tool.	on
square_check_existing_client	Existing Client Lookup	Master on/off for the auto-load-bookings-by-phone behaviour at conversation start.	on
square_setup_scenarios	Auto-Setup Canvas Scenarios	Publishes the consultation/booking/cancel scenarios on first install.	on

Advanced fine-tuning (hidden by default)

There are advanced settings — labelled *Advanced* — *for admins only* in the UI — that let you fine-tune **what the agent extracts** from the conversation (which service, what date, what time, which slot, which phone number) and **how strict the catalog match is**. They're hidden by default. Clear a value and **Publish All** to restore the shipped default.

Attribute	Title in UI	Purpose	Default
Catalog score threshold	AKB Catalog Score Threshold	Minimum fuzzy-match score for a candidate to surface. Lower = more permissive (more candidates), higher = stricter. Raise to 0.3–0.5 if the agent picks the wrong variation on ambiguous names; lower toward 0 if good matches are missed.	0.1
Catalog max topics	AKB Catalog Max Topics	Maximum candidates the AKB returns to the agent's ranking step. Wider = more context, narrower = tighter prompt. Raise for large catalogs (20–30) or lower for small ones.	10
Consultation max services	Consultation Max Services	Caps how many variations the agent reads out in a single consultation reply. Voice channels usually want 2–3; chat channels can handle more.	3
square_gated_services_marker	Gated Services Marker	Marker phrase the integration scans for in the Square service description to flag the service as existing-client-only. Case-insensitive substring match. Set to a clear, unique phrase or wrap it in brackets ([Existing clients only]) for visual clarity in the Square UI. Leave empty to disable the gate.	Only for existing client

Internal settings (OAuth tokens, the resolved Square endpoint, a setup persona ID) are managed automatically and never need manual editing.

Changelog

v2.5.0 — Session-driven AKB sync, NAF procedures, gating polish

- **Catalog sync moved to a session-driven model with TTL + race lock.** Replaces the "rebuild on every publish" model. `CatalogFlow/MaybeSyncCatalogOnSessionSkill` subscribes to `session_started` and runs a fresh sync at most once every 2 hours; `CatalogFlow/UpdateCatalogSkill` enforces the TTL plus an in-progress lock (with 10-minute stale-takeover) on every entry. Newo's project-publish flow wipes AKB topics, so `InitFlow/SetupSkill` clears `square_catalog_last_sync_at` before dispatching a publish-time refresh — that re-populates the AKB immediately on every Publish All. Two new hidden attributes manage the cycle: `square_catalog_last_sync_at` (UTC ISO-8601 of the last successful sync) and `square_catalog_sync_in_progress` (in-flight lock).

- **Defensive guard against an "empty save" race.** If `_saveCatalogToAkbSkill` ever sees an empty `square_catalog_raw` while existing `square_catalog` topics are still in AKB, it refuses to proceed to the orphan-delete step — that combination means another sync chain already saved successfully and proceeding would `DeleteAkb` -wipe every topic. The lock is still released so the next legitimate sync isn't blocked. Closes [#1144](#).
- **Booking scenario rewritten around shared NAF procedures.** The hand-written "ask name → ask phone → ask email" block in step 5.5 is replaced with delegated calls to canonical procedures: `Gathering Preferred Date and Time`, `Reconfirming or Gathering Phone Number`, `Reconfirming or Gathering First and Last Name`. The phone procedure now runs **before** availability so the integration can verify existing-client status by phone before the AKB gating filter strips topics — this fixes the silent block of generic parent terms ("YAG Laser Tattoo Removal") when only specific variations ("Package") were gated.
- **Date/time captured before availability.** Step 7.2 (was: enumerate slots after lookup) now ALWAYS asks the User for preferred date AND time first, then the availability code-phrase fires with the captured preference in context. Eliminates the "agent picks the first slot from the list without asking" hallucination.
- **Catalog AKB cleanup.** Service description is no longer duplicated as a `description` fact — it lives only in `Summary ( ... | Description: ... )`. The existing-client gating filter substring-matches against `Summary` directly. -1 redundant copy of every description text, +1 less brittle parse path.
- **Explicit-block fires only when there are no non-gated alternatives that match.** `_checkAvailabilitySkill` now pre-checks the request against kept (non-gated) topic names: if the request also matches any of them, fall through to silent LLM substitution instead of blocking. Generic parent terms (e.g. "YAG Laser Tattoo Removal" when only "YAG Laser Tattoo Removal Package" is gated) no longer trigger the block.
- **Centralised end-of-session prompt-section cleanup.** `ExistingClientFlow/ClearDataSkill` now removes all five tool-owned prompt sections (`AvailableSlots`, `BookingResult`, `CancellationResult`, `ExistingBookings`, `ExistingCustomerInfo`) on `end_session` instead of only the two it owned. Guards against future regressions where another flow's `ClearDataSkill` silently fails to fire.
- **Canvas scenarios re-numbered to avoid NAF preset collisions.** The NAF beauty preset already uses Scenario 5 (Regular Transfer) and Scenario 6 (Consultation and Lead Qualification); our custom scenarios collided with those. Renamed:
 - "Scheduling Appointment in Square via Agent": 5 → **7**
 - "Canceling Appointment in Square via Agent": 6 → **8**
 - "Service Consultation in Square": 7 → **9**

v2.4.0 — Existing-client gating + chat phone-lookup workaround + slot presentation polish

- **Gate services marked "for existing clients only".** New `square_gated_services_marker` setting (default `Only for existing client`) flags services whose description contains the marker. New callers no longer see those services in availability or consultation results; if they ask for one by name, the agent replies with a polite block message and offers a non-gated alternative. Returning customers identified by phone see the full catalog. Closes [#1046](#).
- **Returning-customer flag.** `ExistingClientFlow/_processClientResultSkill` now writes `"true" / "false"` after every `Square /v2/customers/search`, so other flows can branch on verified existing-client status without re-running the lookup.
- **Chat workaround for phone identification.** Voice channels auto-fire `define_user_phone_number` on conversation start, but chat sessions don't carry caller-id. The

availability skill now extracts the phone from the conversation memory mid-turn, fires the same Square customer search, saves the conversation context to a persona attribute, and re-fires `square_check_availability` once the lookup resolves — so existing clients in chat are unblocked automatically without requiring a second user message.

- **Catalog parser fix — descriptions reach AKB.** `_processCatalogPageSkill` was reading `description` from the wrong nesting level (`object.description` instead of `item_data.description_plaintext`). Every AKB topic was being written with an empty `description` fact, which silently broke the marker-based gating. Now the parser prefers `description_plaintext` with `description` and `description_html` as fallbacks.
- **Sub-case slot presentation.** Booking scenario step 5.3 / Case2 now picks one of three sub-cases based on what the customer said:
 - **Sub-case A** — specific date and time → confirm in one line: "Yes, [date] at [time] is available — would you like me to book it?" . No more enumerating the day's slots after a specific request.
 - **Sub-case B** — only a day named → compact one-line-per-day list, then "Which time works best?".
 - **Sub-case C** — no date → next 1–3 days in compact form, then "Which day and time work best?".
- **Self-cleaning canvas section.** Every early exit in `_checkAvailabilitySkill` (gated, no-match, ambiguous) now writes a fresh `AvailableSlots` section with an explicit status (`not_available_for_new_clients` , `no_match` , `needs_clarification`) — the canvas no longer gets stuck in `In progress` after an early exit, and the booking scenario reacts to each status with the right behaviour.

v2.3.0 — Mi Vida hardening: catalog upsert, disambiguation, advisory consultation

- **Catalog sync is now an upsert.** Replaces the previous wipe-and-rebuild. AKB stays consistent throughout every Publish All — `SearchFuzzyAkb` calls during the refresh always return real results. Topics are created, updated in place, and deleted only after the new snapshot has been written. Orphans (variations no longer in Square) and legacy rows without a parseable `variation_id` are swept on the same pass.
- **Tunable AKB fuzzy match.** New advanced settings let admins dial match strictness without a redeploy:
 - Catalog score threshold (default `0.1`) — minimum fuzzy score for a candidate.
 - Catalog max topics (default `10`) — how many candidates feed the agent's ranking step.
- **Consultation reply cap.** New Consultation max services (default `3`) caps how many variations the agent voices in a single consultation reply, so voice channels don't enumerate ten options.
- **Service disambiguation in availability.** When the catalog returns several plausible matches for the customer's request, the agent no longer guesses: it lists the candidates with optional short hints, asks which one, and re-fires the availability lookup with the clarified service. Booking scenario step 5.3 has a new case for this — operator-visible without code changes.
- **Comparative follow-ups stay in context.** Questions like "which is cheapest?", "what's the longest one?", "what's the difference between 1 session and 3?" are now answered from the data the agent already has, without re-firing the consultation tool. Consultation reply now includes price + duration so this is possible.
- **Advisory consultation queries.** Phrases like "what do you have for legs?", "popular brow options", "something for hair removal" now go to the catalog using the area as a search hint, instead of looping on "tell me a specific service name".
- **Anti-hallucination guard.** Canvas instruction makes the agent strict about only reading out catalog services that came from a tool result. No more invented Profhilo / Dermal Filler suggestions.

v2.2.0 — Existing-client recognition + cross-session and multi-booking cancel

- **New ExistingClientFlow** . When phone is detected on a conversation, the agent searches Square customers, loads the upcoming bookings, populates the persona cache, and emits `<ExistingCustomerInfo>` and `<ExistingBookings>` prompt sections so the canvas can read the booking list directly. Gated by **Existing Client Lookup** .
- **Cross-session cancellation works**. A customer who books today and calls tomorrow is recognised; the agent finds their booking via Square and cancels it without asking for a booking ID.
- **Multi-booking disambiguation**. When the customer has more than one upcoming booking, the agent reads the list back and asks which to cancel. The right one is picked via an LLM extraction validated against the actual `booking_id` list — never guesses.
- **Custom search-bookings tool** (custom booking-search behavior). Replaces the standard NAF `search_booking_tool` so cancellation can be triggered explicitly when needed.
- **Booking flow hardening**. Phone is now extracted from the conversation in E.164 with a 6-step fallback chain; `bookingDateTime` is normalized at write time so the persona cache always carries the canonical `YYYY-MM-DD HH:MM` shape; every bail-out branch clears the `<BookingResult>` / `<CancellationResult>` placeholder so the agent never gets stuck waiting for a result that won't arrive.
- **Canvas Step 5.1 is no longer gated by the consultation match** — the agent always proceeds to live availability lookup once a single service has been named.
- **Catalog the old catalog timestamp removed**. The catalog is rebuilt on every publish; the timestamp attribute is dropped automatically by a migration.

v1.3.0 — Tighter slot presentation + reliability fixes

- **Compact slot presentation**. Up to **three days** with up to **five well-spaced slots per day** instead of dumping every 30-minute interval.
- **Better handling of "next Tuesday" and similar phrases**. Agent stops inventing dates that don't match the day of the week the customer named.
- **Catalog reliably reaches the agent on first publish**.
- **Smoother first-time setup** — the location-actor resolution error is fixed.

v1.2.0 — Service Consultation tool + live pricing + better extraction

- **Service Consultation tool and scenario** for "do you offer X?" / "how much is Y?".
- **Price and duration** stored alongside every variation in the agent's knowledge base.
- **Calendar-aware date / time extraction** for both availability and booking.
- **Forgiving catalog matching** — typos and partial names resolve to the right service.

v1.1.x — earlier Square integration work

Historical versions before the v1.2.0 catalog/consultation overhaul. Full commit history: [branch 912-improvesquareup-hcp-style-akb](#) .